

## LAWSON AAF, GA, USA

WMO: 722250

Lat: 32.332N

Lon: 84.989W

Elev: 71

StdP: 100.48

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13829

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -5.4       | -3.5 | -12.3                         | 1.3 | 0.5  | -9.8 | 1.6 | 1.9  | 9.2                   | 7.6  | 8.3 | 8.3  | 1.9                   | 320  | 0.366 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.9                   | 36.0            | 24.4 | 34.9 | 24.4 | 33.7 | 24.3 | 27.0                | 31.4 | 26.3 | 31.4 | 25.8 | 30.8 | 2.9                  | 300  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 26.1                            | 21.6 | 28.5 | 25.1 | 20.4 | 27.9 | 24.5 | 19.7 | 27.5 | 85.0          | 31.5 | 82.0 | 31.4 | 79.6 | 31.1 | 31.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 7.7               | 6.7  | 5.5 | -9.4                       | 37.5 | 2.0                | 1.8 | -10.8                                              | 38.8 | -11.9      | 39.8 | -13.0      | 40.9 | -14.4      | 42.2 |
|                   |      |     | -9.9                       | 28.4 | 2.0                | 1.1 | -11.3                                              | 29.1 | -12.4      | 29.8 | -13.5      | 30.4 | -15.0      | 31.2 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 18.2   | 7.9   | 10.6  | 14.0  | 17.5  | 22.3  | 26.0  | 27.3  | 27.0  | 24.2  | 18.7  | 12.9  | 9.5  |
|                                                                                  | DBStd       | 7.85   | 5.26  | 5.18  | 4.49  | 3.62  | 3.08  | 2.06  | 1.67  | 1.84  | 2.76  | 4.15  | 4.61  | 5.05 |
|                                                                                  | HDD10.0     | 268    | 106   | 51    | 16    | 1     | 0     | 0     | 0     | 0     | 0     | 1     | 21    | 73   |
|                                                                                  | HDD18.3     | 1252   | 324   | 220   | 146   | 58    | 7     | 0     | 0     | 0     | 1     | 47    | 172   | 277  |
|                                                                                  | CDD10.0     | 3258   | 41    | 68    | 138   | 227   | 380   | 479   | 537   | 526   | 426   | 272   | 108   | 57   |
|                                                                                  | CDD18.3     | 1201   | 1     | 3     | 11    | 34    | 129   | 229   | 278   | 268   | 178   | 59    | 9     | 2    |
|                                                                                  | CDH23.3     | 11864  | 6     | 31    | 156   | 464   | 1376  | 2200  | 2774  | 2601  | 1596  | 563   | 86    | 12   |
| CDH26.7                                                                          | 5361        | 0      | 2     | 22    | 116   | 558   | 1077  | 1407  | 1299  | 714   | 156   | 11    | 0     |      |
| Wind                                                                             | WSAvg       | 2.1    | 2.4   | 2.5   | 2.6   | 2.3   | 2.0   | 1.7   | 1.7   | 1.7   | 1.9   | 1.9   | 2.0   | 2.1  |
| Precipitation                                                                    | PrecAvg     | 1257   | 111   | 124   | 146   | 104   | 82    | 105   | 132   | 115   | 83    | 68    | 97    | 110  |
|                                                                                  | PrecMax     | 1717   | 208   | 377   | 329   | 250   | 234   | 197   | 278   | 228   | 273   | 193   | 278   | 410  |
|                                                                                  | PrecMin     | 894    | 50    | 30    | 14    | 19    | 12    | 22    | 66    | 52    | 14    | 12    | 26    | 45   |
|                                                                                  | PrecStd     | 222    | 33    | 64    | 78    | 53    | 49    | 51    | 55    | 47    | 56    | 51    | 60    | 71   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 24.3  | 26.3  | 29.0  | 31.1  | 34.4  | 36.8  | 37.4  | 37.9  | 36.0  | 32.1  | 27.9  | 25.1 |
|                                                                                  |             | MCWB   | 17.7  | 18.8  | 18.8  | 20.3  | 22.4  | 24.0  | 24.4  | 25.0  | 23.3  | 22.7  | 19.9  | 19.3 |
|                                                                                  | 2%          | DB     | 21.9  | 24.0  | 27.0  | 29.5  | 32.7  | 35.2  | 35.7  | 35.8  | 34.1  | 30.0  | 25.9  | 22.7 |
|                                                                                  |             | MCWB   | 16.9  | 17.7  | 17.9  | 19.9  | 22.3  | 24.2  | 24.8  | 24.8  | 22.9  | 21.3  | 18.7  | 18.5 |
|                                                                                  | 5%          | DB     | 19.8  | 22.1  | 25.1  | 28.0  | 31.3  | 33.9  | 34.3  | 34.3  | 32.5  | 28.3  | 23.8  | 20.5 |
|                                                                                  |             | MCWB   | 16.2  | 16.7  | 16.8  | 19.2  | 21.8  | 24.1  | 24.9  | 24.8  | 23.1  | 20.3  | 17.5  | 17.4 |
|                                                                                  | 10%         | DB     | 17.5  | 20.0  | 22.9  | 26.3  | 30.0  | 32.5  | 33.0  | 32.9  | 31.0  | 26.8  | 22.0  | 18.5 |
|                                                                                  |             | MCWB   | 14.5  | 15.5  | 15.6  | 18.3  | 21.3  | 24.0  | 24.9  | 24.8  | 23.0  | 19.7  | 17.3  | 15.9 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 20.0  | 21.1  | 21.0  | 23.3  | 24.9  | 27.1  | 28.9  | 29.1  | 26.3  | 25.1  | 22.5  | 21.5 |
|                                                                                  |             | MCDB   | 21.6  | 23.9  | 24.5  | 27.2  | 29.2  | 31.8  | 31.7  | 32.0  | 29.7  | 27.7  | 24.6  | 23.0 |
|                                                                                  | 2%          | WB     | 18.5  | 19.4  | 19.9  | 21.9  | 24.0  | 26.3  | 27.0  | 27.1  | 25.4  | 23.8  | 21.0  | 19.9 |
|                                                                                  |             | MCDB   | 20.4  | 22.1  | 23.7  | 26.2  | 28.8  | 31.8  | 31.3  | 31.6  | 30.0  | 27.3  | 23.1  | 21.2 |
|                                                                                  | 5%          | WB     | 16.9  | 18.2  | 18.8  | 20.9  | 23.3  | 25.6  | 26.3  | 26.3  | 24.7  | 22.7  | 19.5  | 18.2 |
|                                                                                  |             | MCDB   | 18.9  | 20.8  | 23.1  | 25.2  | 28.5  | 31.0  | 31.3  | 31.4  | 29.3  | 25.8  | 22.3  | 19.8 |
|                                                                                  | 10%         | WB     | 15.2  | 16.8  | 17.5  | 20.0  | 22.7  | 25.0  | 25.7  | 25.7  | 24.0  | 21.5  | 17.9  | 16.5 |
|                                                                                  |             | MCDB   | 17.1  | 19.2  | 21.0  | 24.2  | 27.9  | 30.1  | 30.9  | 30.5  | 28.4  | 24.5  | 20.8  | 18.4 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 13.7  | 14.2  | 14.8  | 15.4  | 14.3  | 12.4  | 11.9  | 11.9  | 12.5  | 14.2  | 14.9  | 13.0 |
|                                                                                  |             | MCDBR  | 15.5  | 15.6  | 16.9  | 17.0  | 15.5  | 14.6  | 13.6  | 13.9  | 14.8  | 15.4  | 16.2  | 14.4 |
|                                                                                  |             | MCWBR  | 10.6  | 9.5   | 8.5   | 8.2   | 6.4   | 5.3   | 4.6   | 4.8   | 5.5   | 7.3   | 9.3   | 9.9  |
|                                                                                  | 5% WB       | MCDBR  | 12.6  | 12.7  | 13.6  | 13.4  | 13.0  | 12.1  | 11.9  | 11.7  | 11.6  | 11.2  | 12.2  | 11.6 |
|                                                                                  |             | MCWBR  | 9.8   | 9.1   | 7.9   | 7.6   | 6.0   | 5.4   | 5.0   | 5.0   | 5.1   | 6.5   | 8.6   | 9.1  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.327  | 0.329 | 0.357 | 0.389 | 0.440 | 0.485 | 0.521 | 0.523 | 0.428 | 0.376 | 0.352 | 0.330 |      |
|                                                                                  | taud        | 2.518  | 2.506 | 2.446 | 2.345 | 2.244 | 2.174 | 2.082 | 2.069 | 2.325 | 2.446 | 2.460 | 2.521 |      |
|                                                                                  | Ebn at Noon | 896    | 929   | 923   | 901   | 854   | 811   | 780   | 773   | 840   | 864   | 856   | 872   |      |
|                                                                                  | Edh at Noon | 86     | 95    | 109   | 125   | 140   | 150   | 163   | 163   | 121   | 100   | 90    | 81    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.86   | 3.43  | 4.55  | 5.67  | 6.27  | 6.12  | 6.03  | 5.49  | 4.85  | 4.16  | 3.20  | 2.45  |      |
|                                                                                  | RadStd      | 0.16   | 0.40  | 0.38  | 0.41  | 0.47  | 0.48  | 0.37  | 0.33  | 0.43  | 0.51  | 0.28  | 0.26  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only            | +0.43 | N/A     | N/A    | N/A     | +0.43 | +0.61 | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (42 neighbors) | +0.48 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +159    | +107    |

Nomenclature: See separate page